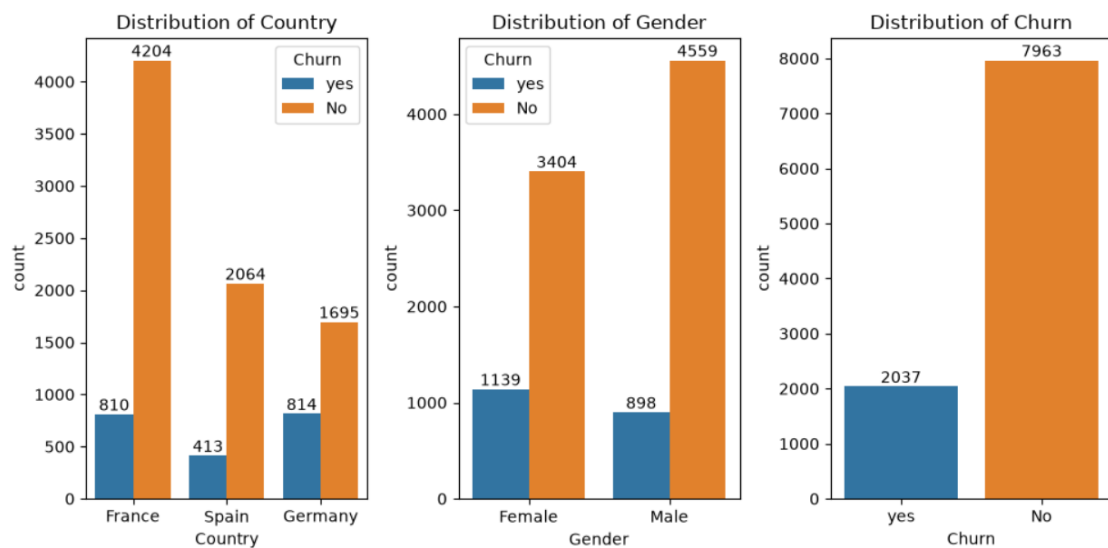


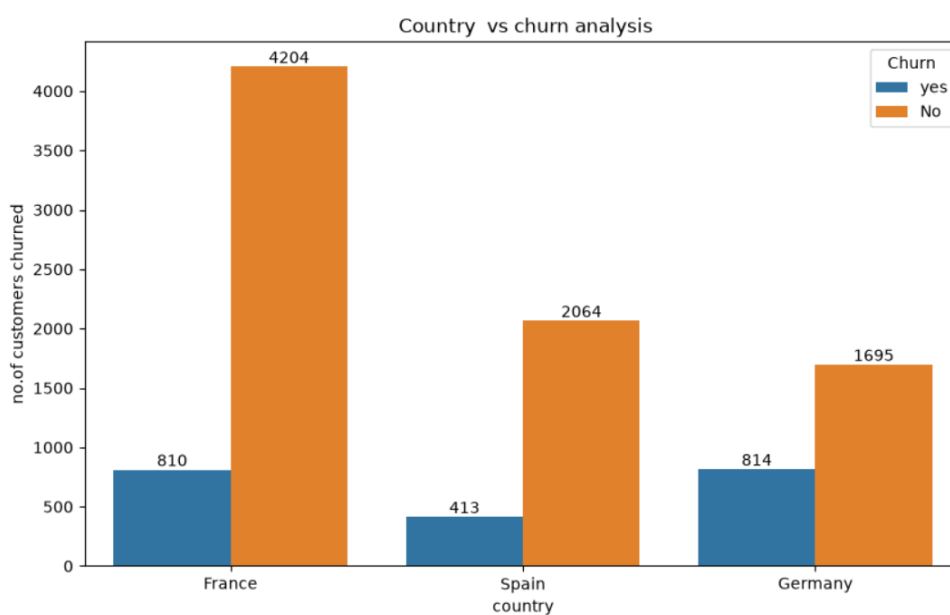
1.Dataset

	customer_id	credit_score	country	gender	age	tenure	balance	products_number	credit_card	active_member	estimated_salary	churn
0	15634602	619	France	Female	42	2	0.00	1	1	1	101348.88	1
1	15647311	608	Spain	Female	41	1	83807.86	1	0	1	112542.58	0
2	15619304	502	France	Female	42	8	159660.80	3	1	0	113931.57	1
3	15701354	699	France	Female	39	1	0.00	2	0	0	93826.63	0
4	15737888	850	Spain	Female	43	2	125510.82	1	1	1	79084.10	0

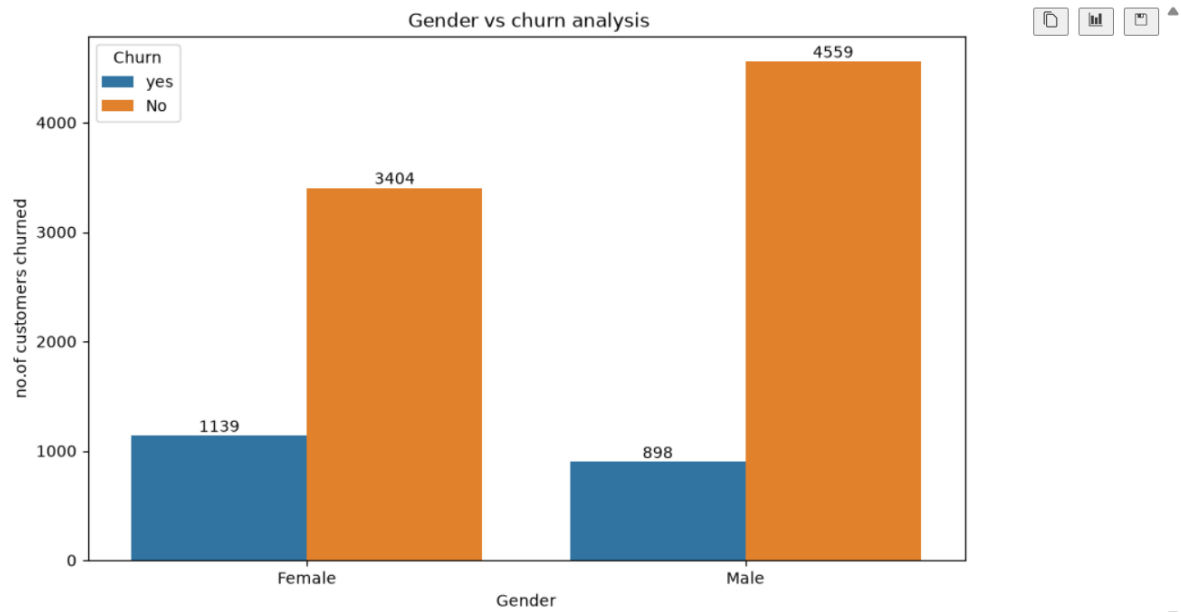
2. Univariate analysis of categorical columns



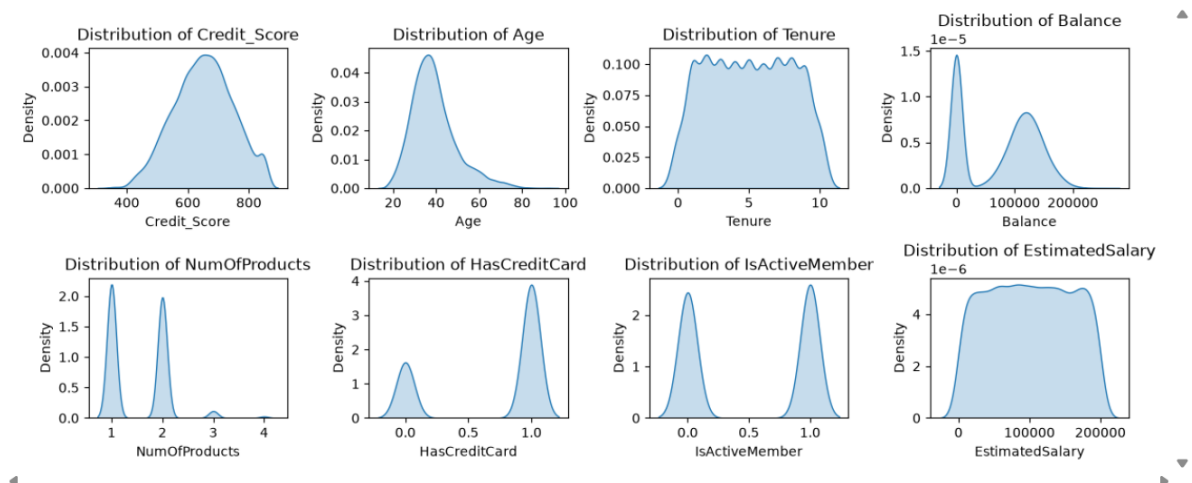
3. Country vs Churn



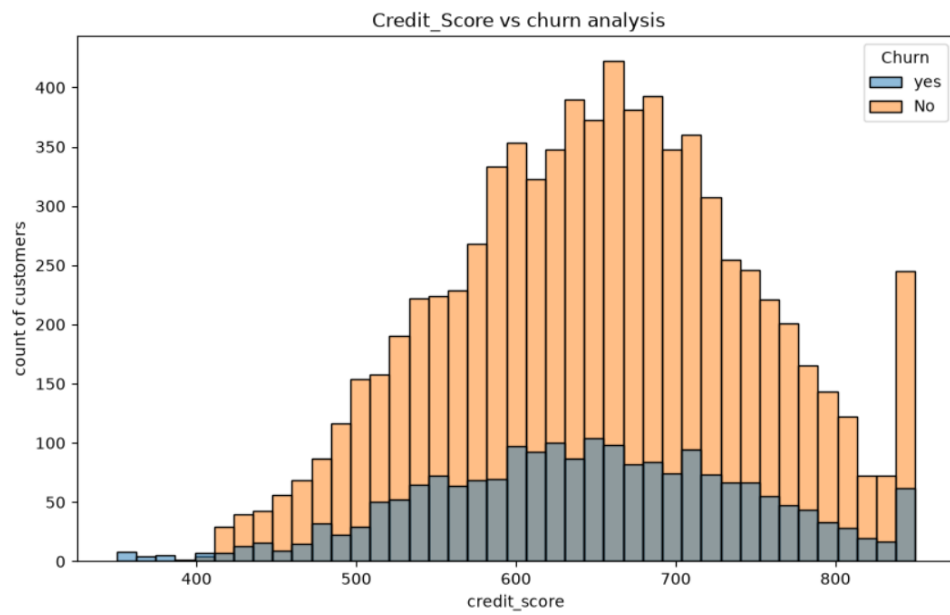
4. Gender vs Churn



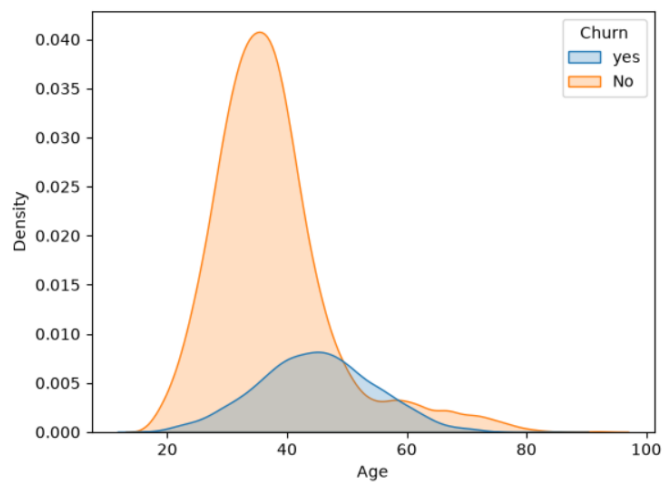
5. Univariate analysis of Numerical Columns



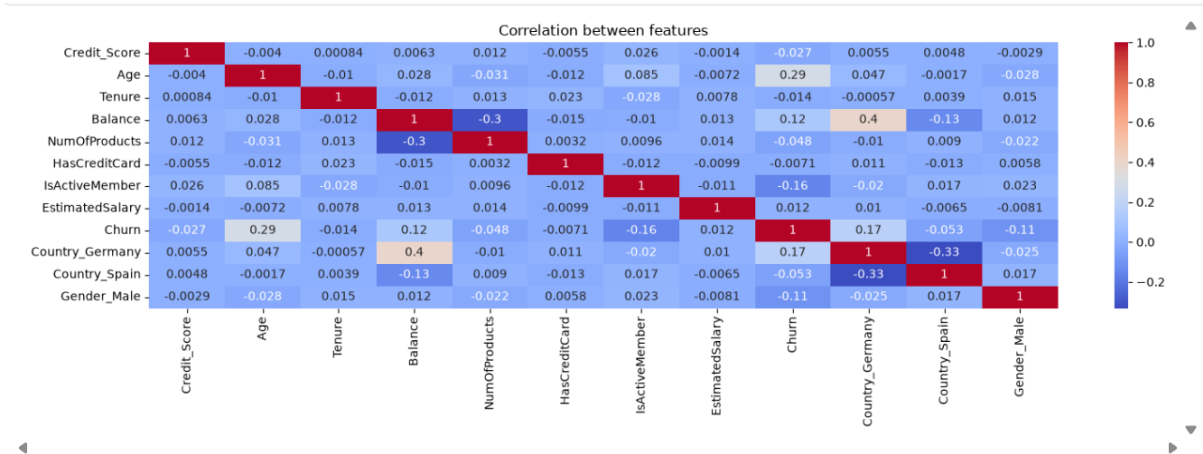
6. Credit score vs Churn



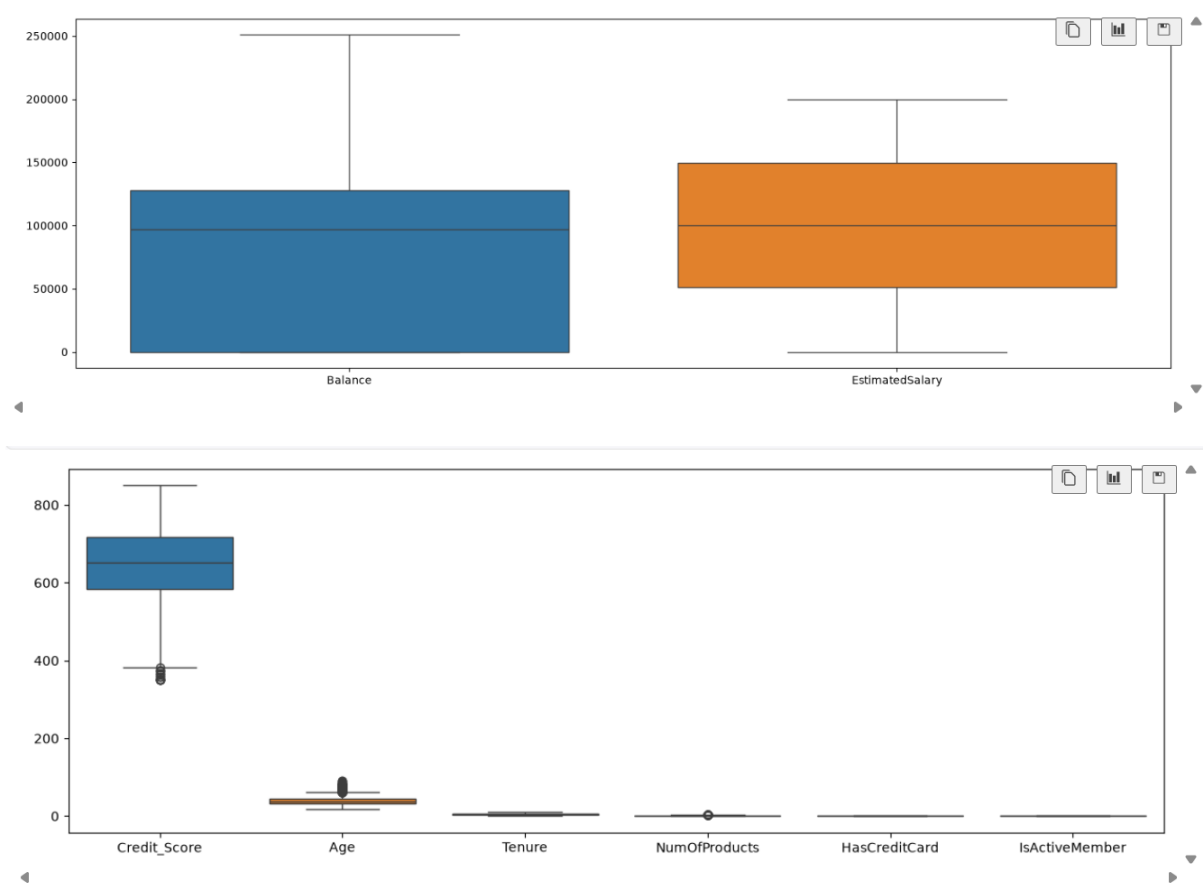
7. Age vs Churn



8.HeatMap



9. Outliers



Observations:

Model	Accuracy	Churn Precision	Churn Recall	Churn F1-Score	Status / Conclusion
Logistic Regression	0.71	0.38	0.71	0.49	High Recall, but lower Precision
KNeighborsClassifier	0.84	0.66	0.44	0.53	Good Accuracy, but misses churners (Low Recall)
RandomForestClassifier	0.85	0.62	0.59	0.61	Balanced performance (Top Candidate)
SVM Classifier (SVC)	0.86	0.8	0.38	0.51	High Precision, but very low Recall
DecisionTreeClassifier	0.79	0.48	0.5	0.49	Moderate performance across all metrics
GaussianNB	0.83	0.63	0.36	0.46	Lowest Churn Recall
XGBClassifier	0.83	0.57	0.64	0.6	High Recall & Strong F1-Score (Top Candidate)

Model Stage	Accuracy	Churn Precision	Churn Recall	Churn F1-Score	Key Improvement
Baseline Random Forest	0.85	0.62	0.59	0.61	Initial benchmark model
Tuned Random Forest	0.84	0.6	0.64	0.62	Higher Recall (+5%) & Better F1-Score

Model Scenario	Accuracy	Churn Precision (Class 1)	Churn Recall (Class 1)	Churn F1-Score (Class 1)	True Positives (Caught Churns)
Tuned Random Forest (No SMOTE)	84.36%	0.6	0.64 (64%)	0.62	316 / 497
Tuned Random Forest (+ SMOTE)	83.00%	0.57	0.61 (61%)	0.59	304 / 497

Metric	Tuned Random Forest	Tuned XGBoost	Best Choice
Accuracy	84.44%	83.44%	RF (Slightly higher overall)
F1 Score	0.6182	0.6208	XGBoost
Recall (Class 1 - Churn)	0.63 (63%)	0.68 (68%)	XGBoost (Key Metric)
True Positives (Detected Churns)	315 / 497	339 / 497	XGBoost (+24 extra churners caught)
False Negatives (Missed Churns)	182	158	XGBoost (Lower risk)

10. Sample Request

← → ↻ http://127.0.0.1:8000/docs#/default/predict_churn_predict_post

GET / Home

POST /predict Predict Churn

Parameters Cancel Reset

No parameters

Request body required application/json

Edit Value | Schema

```
{
  "Credit_Score": 500,
  "Age": 78,
  "Tenure": 1,
  "Balance": 5632,
  "NumOfProducts": 1,
  "HasCreditCard": 0,
  "IsActiveMember": 1,
  "EstimatedSalary": 58695,
  "Country": "Germany",
  "Gender": "Male"
}
```

Execute Clear

11. Sample Response

← → ↻ http://127.0.0.1:8000/docs#/default/predict_churn_predict_post

Responses

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/predict' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
    "Credit_Score": 500,
    "Age": 78,
    "Tenure": 1,
    "Balance": 5632,
    "NumOfProducts": 1,
    "HasCreditCard": 0,
    "IsActiveMember": 1,
    "EstimatedSalary": 58695,
    "Country": "Germany",
    "Gender": "Male"
  }'
```

Request URL

http://127.0.0.1:8000/predict

Server response

Code	Details
200	Response body <pre>{ "prediction": 0, "status": "Not churned" }</pre> Download Response headers <pre>content-length: 39 content-type: application/json date: Thu, 23 Jul 2026 18:59:56 GMT server: uvicorn</pre>